

10.1.3 Network Communication

In CANopen, the communication objects transmit data, trigger events, signal errors, etc. Each communication object is assigned a unique COB-ID (Communication Object Identifier) in the network.

The COB-ID based on device profile DS-401 is assigned according to the following table.

Table 62: COB-ID Assignment

10	9	8	7	6	5	4	3	2	1	0
Function code				Node identifier (0[= all], 1-127)						

Table 63: Communication Objects

Communication Objects		Function Code		Resulting COB-ID		Object Directory Index
		dec	bin	dec	bin	hex
Broadcast messages	NMT commands	0	0000	0		
	Sync message	1	0001	128	80	1005, 1006, 1007
	System time	2	0010	256	100	1012, 1013
Peer-to-Peer messages	Alarm objects	1	0001	129-255	81-FF	1014, 1015
	Tx-PDO1	3	0011	385-511	181-1FF	1800
	Rx-PDO1	4	0100	513-639	201-27F	1400
	Tx-PDO2	5	0101	641-767	281-2FF	1801
	Rx-PDO2	6	0110	769-895	301-37F	1401
	Tx-PDO3	7	0111	897-1023	381-3FF	1802
	Rx-PDO3	8	1000	1025-1151	401-47F	1402
	Tx-PDO4	9	1001	1153-1279	481-4FF	1803
	Rx-PDO4	10	1010	1281-1407	501-57F	1403
	Tx-SDO	11	1011	1409-1535	581-5FF	1200
	Rx-SDO	12	1100	1537-1663	601-67F	1200
	Node monitoring	14	1110	1793-1919	701-77F	100C, 100D 1016, 1017

The parameters required for the communication objects and the parameters/data of the CANopen devices are stored in the object directory.

Which and how many objects are supported by the node depends on the respective fieldbus coupler/controller.

Apart from some specific objects, e.g., for network management (NMT), synchronization (SYNC) or error messages (EMCY), the communication profile contains the two object types PDO and SDO.

The PDOs (Process Data Objects) are used to transmit real-time data and the SDOs (Service Data Objects) allow read/write access to the object directory.

10.1.3.1 Communication Objects

10.1.3.1.1 Process Data Object - PDO

PDOs contain, e.g., real-time data with high-priority identifiers. The data telegrams consist of a maximum of 8 bytes. They can be freely exchanged between the individual modules. Data exchange can be event-driven or synchronized.